Nonlinear Control System – Assignment 4 – Xinyi Yang

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Problem 1

Consider the system

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = a\cos x_1 + u,$$

where $a \in [0.1, 0.5]$. The objective is to design a continuously implemented sliding mode controller that satisfies the control constraint $|u| \le 1$ and stabilizes the system to the origin with steady-state error $|x_1| \le 0.1$.

Choose a linear sliding surface

$$s = cx_1 + x_2, \qquad c > 0.$$

On the sliding surface $s = 0$ we have $x_2 = -cx_1$, so the reduced-order dynamics $\dot{x}_1 = -cx_1$ are exponentially stable.

To avoid chattering we replace $\operatorname{sgn}(s)$ by the saturation function:

$$u = -cx_2 - \hat{a}\cos x_1 - K\,\operatorname{sat}\left(\frac{s}{\varepsilon}\right),$$

where

$$\operatorname{sat}(y) = \begin{cases} y, & |y| \le 1, \\ \operatorname{sgn}(y), & |y| > 1. \end{cases}$$

Take the midpoint of the parameter interval: $\hat{a} = 0.3$, so

$$\Delta_{\max} = \max_{a\in[0.1,0.5]} |a - \hat{a}| = 0.2.$$

Choose $K = 0.25 > \Delta_{\max}$ to guarantee reachability.

Outside the boundary layer the saturation is saturated ($|\operatorname{sat}| = 1$), so

$$|u| \le c|x_2| + \hat{a} + K = c|x_2| + 0.55.$$

In order to satisfy $|u| \le 1$ we need $c|x_2| \le 0.45$. Choosing $c = 0.1$ gives the admissible region $|x_2| \le 4.5$.

Inside the boundary layer $\operatorname{sat}(s/\varepsilon) = s/\varepsilon$, and the $\dot{s}$ equation reduces to

$$\dot{s} = (a - \hat{a})\cos x_1 - \frac{K}{\varepsilon}\,s.$$

Although $K/\varepsilon = 25$ is numerically large, the constraint is not tightened: since $|s| \le \varepsilon$ inside the layer we have $|Ks/\varepsilon| \le K = 0.25$, so

$$|u| \le c|x_2| + \hat{a} + \frac{K}{\varepsilon}|s| \le c|x_2| + \hat{a} + K \le 1 \qquad (|x_2| \le 4.5).$$

Hence the constraint $|u| \le 1$ is not violated inside the boundary layer either.

At steady state $\dot{s} = 0$, giving

$$s \approx \frac{\varepsilon}{K}(a - \hat{a})\cos x_1, \qquad |s| \le \frac{0.2\,\varepsilon}{K}.$$

Because $s = cx_1 + x_2$, we have $\dot{x}_1 = x_2 = s - cx_1$. Setting $\dot{x}_1 \approx 0$ at steady state yields $x_1 \approx s/c$, hence

$$|x_1| \le \frac{0.2\,\varepsilon}{cK}.$$

The requirement $|x_1| \le 0.1$ yields

$$\varepsilon \le \frac{0.1\,cK}{0.2} = 0.5\,cK.$$

With $c = 0.1$ and $K = 0.25$: $\varepsilon \le 0.5 \times 0.1 \times 0.25 = 0.0125$. We choose $\varepsilon = 0.01$.

Final controller.

$$u = -0.1\,x_2 - 0.3\cos x_1 - 0.25\,\mathrm{sat}\left(\frac{x_2 + 0.1x_1}{0.01}\right).$$

- Control constraint: $|u| \le 1$ provided $|x_2| \le 4.5$.
- Convergence: The system is driven onto the boundary layer $|s| \le \varepsilon$ in finite time and remains there; subsequently the trajectory converges to the origin exponentially along the sliding surface $\dot{x}_1 = -cx_1$.
- Steady-state error: $|x_1| \le \dfrac{0.2 \times 0.01}{0.1 \times 0.25} = 0.08 < 0.1$.

Remark on numerical implementation. The small boundary layer thickness $\varepsilon = 0.01$ requires a fine integration step in simulation to resolve the linear region of the saturation. In practice one may increase ε (with a corresponding reduction of c or increase of K to keep the error bound satisfied) if numerical stiffness is a concern.

Problem 2

Consider the system

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = -x_1^3 + u, \quad y = x_1.$$

Choose

$$u = x_1^3 + v \quad \Longrightarrow \quad \dot{x}_1 = x_2,\ \dot{x}_2 = v.$$

Select $v = -k_1x_1 - k_2x_2$ with $k_1 = 1$, $k_2 = 2$ (eigenvalues at -1, repeated):

$$u = \alpha(x_1, x_2) = x_1^3 - x_1 - 2x_2.$$

Rewrite the plant in observer form

$$\dot{x} = Ax + \psi(u, y), \quad y = Cx,$$

with

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad \psi(u, y) = \begin{pmatrix} 0 \\ -y^3 + u \end{pmatrix}.$$

The key observation is that ψ depends *only* on the known signals u and $y = x_1$, not on the unmeasured state x_2. This is what makes the observer error dynamics linear (see below). The pair (A, C) is observable.

By (11.17) the observer is

$$\dot{\hat{x}} = A\hat{x} + \psi(u, y) + H(y - C\hat{x}),$$

i.e.

$$\dot{\hat{x}}_1 = \hat{x}_2 + h_1(x_1 - \hat{x}_1), \qquad \dot{\hat{x}}_2 = -x_1^3 + u + h_2(x_1 - \hat{x}_1).$$

Choose $H = (h_1, h_2)^\top$ so that $A - HC$ is Hurwitz. With $h_1 = 4$, $h_2 = 4$ the characteristic polynomial of $A - HC$ is $s^2 + 4s + 4 = (s+2)^2$. Thus

$$H = \begin{pmatrix} 4 \\ 4 \end{pmatrix}, \quad \text{eigenvalues at } s = -2 \text{ (repeated).}$$

Define $\tilde{x} = x - \hat{x}$. Subtracting the observer equation from the plant equation, and using the fact that $\psi(u, y)$ is *identical* in both (since it depends only on the known signals u and y, not on $\hat{x}$), the nonlinear terms cancel exactly:

$$\dot{\tilde{x}} = A\tilde{x} - HC\tilde{x} = (A - HC)\tilde{x}.$$

This linear equation is globally exponentially stable (GES).

Replace x_2 by $\hat{x}_2$ in the state-feedback law:

$$u = \alpha(x_1, \hat{x}_2) = y^3 - y - 2\hat{x}_2.$$

Writing $\hat{x}_2 = x_2 - \tilde{x}_2$ the closed-loop plant becomes

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = -x_1 - 2x_2 + 2\tilde{x}_2,$$

i.e.

$$\dot{x} = A_c x + B\tilde{x}_2, \quad A_c = \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$$

The overall system is a cascade:

$$\begin{aligned} \dot{x} &= A_c x + B\tilde{x}_2, \\ \dot{\tilde{x}} &= (A - HC)\tilde{x}. \end{aligned}$$

- Nominal subsystem ($\tilde{x}_2 = 0$): $\dot{x} = A_c x$ is globally asymptotically stable (GAS) since A_c is Hurwitz.

- ISS of the nominal subsystem: For the linear system $\dot{x} = A_c x + B\tilde{x}_2$, let P_c satisfy $P_c A_c + A_c^\top P_c = -I$. Then $V_x = x^\top P_c x$ satisfies $\dot{V}_x \leq -\|x\|^2 + 2\|P_c B\|\|x\|\,|\tilde{x}_2|$, which gives the ISS estimate $\dot{V}_x \leq -\frac{1}{2}\|x\|^2$ whenever $\|x\| \geq 4\|P_c B\|\,|\tilde{x}_2|$. Hence the nominal subsystem is ISS with respect to the input $\tilde{x}_2$.

- Driven subsystem: $\dot{\tilde{x}} = (A - HC)\tilde{x}$ is GES, so $\tilde{x}_2(t) \to 0$ exponentially for all initial conditions.

By the third bullet point of Theorem 12.1, the origin of the overall (plant + observer) system is globally asymptotically stable.

Summary

$$\dot{\hat{x}}_1 = \hat{x}_2 + 4(y - \hat{x}_1),$$
$$\dot{\hat{x}}_2 = -y^3 + u + 4(y - \hat{x}_1),$$
$$u = y^3 - y - 2\hat{x}_2, \qquad y = x_1.$$

Problem 3

The system (A.12) with $\varepsilon = 1$ is

$$\dot{z}_1 = z_2, \qquad \dot{z}_2 = -z_1 - h(z_2) + u, \quad h(z_2) = -k_1 z_2 + k_2 z_2^3,$$

with $0.5 \le k_1 \le 1$, $1 \le k_2 \le 2$.

With $u = 0$ the linearization at the origin gives $A = \begin{bmatrix} 0 & 1 \\ -1 & k_1 \end{bmatrix}$ with characteristic polynomial $s^2 - k_1 s + 1$. Since $k_1 \ge 0.5 > 0$, the origin is unstable (eigenvalues have positive real part); the system exhibits limit-cycle oscillations.

At $z = 0$: $u = 0$, $\dot{z}_1 = 0$, $\dot{z}_2 = 0 - h(0) + 0 = 0$. Hence the origin is indeed an equilibrium of the closed-loop system for any controller u satisfying $u(0) = 0$.

The control input enters $\dot{z}_2$ linearly. We use the nominal parameter values

$$\hat{k}_1 = 0.75, \qquad \hat{k}_2 = 1.5$$

(midpoints of the respective intervals) and choose

$$u = \hat{h}(z_2) + v = -\hat{k}_1 z_2 + \hat{k}_2 z_2^3 + v,$$

where v is to be designed. We remark that the controller requires knowledge of $\hat{k}_1$ and $\hat{k}_2$ as nominal values; robustness to the resulting mismatch is analysed below.

Under exact cancellation ($k_i = \hat{k}_i$) this gives $\dot{z}_2 = -z_1 + v$. We stabilize by $v = -c_1 z_1 - c_2 z_2$ with $c_1 = 1$, $c_2 = 2$, yielding the closed-loop polynomial $s^2 + 2s + 2$ (roots $-1 \pm i$). The resulting implementable controller is

$$u = -z_1 - 2.75\, z_2 + 1.5\, z_2^3.$$

This is a polynomial, hence smooth and locally Lipschitz on $\mathbb{R}^2$. Under exact cancellation the nominal closed-loop system $\dot{z}_1 = z_2$, $\dot{z}_2 = -z_1 - 2z_2$ is globally exponentially stable, so $z_1(t) \to 0$ with zero steady-state error.

Let $\tilde{k}_1 = k_1 - \hat{k}_1$ and $\tilde{k}_2 = k_2 - \hat{k}_2$. The actual closed-loop dynamics are

$$\dot{z}_1 = z_2, \qquad \dot{z}_2 = -z_1 - 2z_2 + \Delta(z_2), \quad \Delta(z_2) = \tilde{k}_1 z_2 - \tilde{k}_2 z_2^3,$$

with $|\tilde{k}_1| \le 0.25$ and $|\tilde{k}_2| \le 0.5$.

Consider

$$V(z) = z_1^2 + z_1 z_2 + z_2^2.$$

The associated symmetric matrix is $P = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix}$ with eigenvalues $\frac{1}{2}$ and $\frac{3}{2}$, so V is positive definite and

$$\tfrac{1}{2}\|z\|^2 \le V(z) \le \tfrac{3}{2}\|z\|^2.$$

A direct computation along the nominal system ($\dot{z}_2 = -z_1 - 2z_2$) gives

$$(2z_1 + z_2)z_2 + (z_1 + 2z_2)(-z_1 - 2z_2) = -2(z_1^2 + z_1 z_2 + z_2^2) = -2V,$$

so $\dot{V}\big|_{\Delta=0} = -2V$.

$$\dot{V} = -2V + (z_1 + 2z_2)\Delta(z_2).$$

On the compact set $\|z\| \le r$:

$$|\Delta(z_2)| \le (0.25 + 0.5r^2)|z_2|, \qquad |z_1 + 2z_2| \le \sqrt{5}\,\|z\|.$$

Hence

$$|(z_1 + 2z_2)\Delta| \le \sqrt{5}(0.25 + 0.5r^2)\|z\|^2.$$

Using $\|z\|^2 \le 2V$ (since $\lambda_{\min} = 1/2$, $\|z\|^2 \le V/(1/2) = 2V$):

$$|(z_1 + 2z_2)\Delta| \le 2\sqrt{5}(0.25 + 0.5r^2)\,V \triangleq M(r)\,V,$$

where now $M(r) = 2\sqrt{5}(0.25 + 0.5r^2)$.

Setting $r = 0.1$ gives

$$M(0.1) = 2\sqrt{5}(0.25 + 0.005) = 2\sqrt{5} \times 0.255 \approx 2.236 \times 0.51 \approx 1.14 < 2,$$

so $\dot{V} \le -(2 - 1.14)V = -0.86\,V < 0$ on $\|z\| \le 0.1$.

More generally, $M(r) < 2$ requires

$$2\sqrt{5}(0.25 + 0.5r^2) < 2 \quad\Longleftrightarrow\quad r^2 < \frac{2 - 2\sqrt{5} \times 0.25}{2\sqrt{5} \times 0.5} = \frac{2 - \sqrt{5}/2}{\sqrt{5}} \approx \frac{2 - 1.118}{2.236} \approx 0.394,$$

i.e. $r < r^* \approx 0.628$. We choose $r = 0.5$, giving

$$M(0.5) = 2\sqrt{5}(0.25 + 0.125) = 2\sqrt{5} \times 0.375 \approx 1.677 < 2,$$

so $\dot{V} \le -(2 - 1.677)V \approx -0.32\,V < 0$ on $\|z\| \le 0.5$.

The origin is *locally asymptotically stable* with an attraction region containing at least the ball $\|z\| \le 0.5$. For any initial condition $\|z(0)\| \le 0.5$, the trajectory satisfies $z(t) \to 0$ as $t \to \infty$, and in particular $z_1(t) \to 0$ with zero steady-state error.

Summary

The designed locally Lipschitz controller

$$u = -z_1 - 2.75\,z_2 + 1.5\,z_2^3$$

globally stabilizes the nominal system and, for all initial conditions in the ball $\|z(0)\| \le 0.5$, guarantees zero steady-state error for z_1 in the presence of bounded parameter uncertainties ($|\tilde{k}_1| \le 0.25$, $|\tilde{k}_2| \le 0.5$).

Problem 4

The magnetic levitation system (A.29) with $\varepsilon = 1$ is

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = -bx_2 + 1 + cu,$$

with $b \in [0, 0.1]$, $c \in [0.8, 1.2]$. The goal is to regulate $x_1 \to r \in [0.5, 1.5]$ with zero steady-state error and $-2 \le u \le 0$.

Define $e_1 = x_1 - r$, $e_2 = x_2$. Then

$$\dot{e}_1 = e_2, \qquad \dot{e}_2 = -be_2 + 1 + cu.$$

At steady state $e_1 = e_2 = 0$ and

$$u_{ss} = -\frac{1}{c} \in [-1.25,\ -0.833] \subset [-2, 0].\checkmark$$

Setting $u = -1 + v$ (nominal cancellation of the constant "1" using $\hat{c} = 1$) yields

$$\dot{e}_2 = -be_2 + (1 - c) + cv.$$

The residual constant $(1 - c) \neq 0$ for $c \neq 1$ causes a steady-state bias under pure proportional–derivative feedback; integral action is therefore necessary to achieve zero steady-state error for all $c \in [0.8, 1.2]$.

Introduce

$$\sigma = \int_0^t e_1(\tau)\, d\tau, \qquad \sigma(0) = 0.$$

Design

$$u = -1 + v, \qquad v = -c_1 e_1 - c_2 e_2 - c_3 \sigma,$$

where $c_1, c_2, c_3 > 0$. The augmented (σ, e_1, e_2) system has the nominal closed-loop matrix (using $\hat{b} = 0.05$, $\hat{c} = 1$)

$$A_{cl} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -c_3 & -c_1 & -(\hat{b} + c_2) \end{pmatrix}$$

with characteristic polynomial

$$p(s) = s^3 + (\hat{b} + c_2)s^2 + c_1 s + c_3.$$

Placing all poles at $s = -1$ gives the desired polynomial $(s + 1)^3 = s^3 + 3s^2 + 3s + 1$, so matching coefficients yields

$$c_2 = 3 - \hat{b} = 2.95, \qquad c_1 = 3, \qquad c_3 = 1.$$

At equilibrium $\dot{\sigma} = \dot{e}_1 = \dot{e}_2 = 0$ gives $e_1 = 0$, $e_2 = 0$, and $\sigma = (1 - c)/(cc_3)$ (bounded for all admissible c). Hence the integral action enforces $e_1 = 0$ at every equilibrium, guaranteeing zero steady-state error.

Final controller

$$u = -1 - 3(x_1 - r) - 2.95\, x_2 - \int_0^t (x_1(\tau) - r)\, d\tau.$$

This is a linear function of (e_1, e_2, σ) and hence globally (in particular locally) Lipschitz.

At equilibrium $e_1 = e_2 = 0$ and $\sigma = (1 - c)/c$, so

$$u_{ss} = -1 - \frac{1 - c}{c} = -\frac{1}{c} \in [-1.25,\ -0.833] \subset [-2, 0].\checkmark$$

During transients the unsaturated control may temporarily leave $[-2, 0]$. To strictly enforce the hard constraint we implement

$$u = \mathrm{sat}_{[-2,0]}(-1 - 3e_1 - 2.95e_2 - \sigma)\,.$$

The saturation function is globally Lipschitz; the modified controller remains locally Lipschitz. In a sufficiently small neighbourhood of the equilibrium the saturation is inactive, so the local stability and zero steady-state error properties are preserved.

For actual parameters $b \in [0, 0.1]$ and $c \in [0.8, 1.2]$, the closed-loop characteristic polynomial is

$$q(s) = s^3 + (b + 2.95c)s^2 + 3c\,s + c.$$

All coefficients are positive. For a third-order polynomial $s^3 + a_2 s^2 + a_1 s + a_0$ the Routh–Hurwitz conditions reduce to (i) all coefficients positive, and (ii) $a_2 a_1 > a_0$. We verify both conditions at all four extreme corners of the parameter box:

b	c	$a_2 = b + 2.95c$	$a_1 = 3c$	$a_0 = c$	$a_2 a_1$	$a_2 a_1 > a_0$?
0	0.8	2.36	2.40	0.80	5.664	✓
0.1	0.8	2.46	2.40	0.80	5.904	✓
0	1.2	3.54	3.60	1.20	12.744	✓
0.1	1.2	3.64	3.60	1.20	13.104	✓

In every case $a_2 a_1 \gg a_0$, confirming stability for all admissible (b, c). The minimum of $a_2 a_1 - a_0$ over the parameter box is $5.664 - 0.80 = 4.864 > 0$, achieved at $b = 0$, $c = 0.8$. The integral action guarantees zero steady-state error.

Summary

The designed locally Lipschitz state-feedback controller is

$$u = -1 - 3(x_1 - r) - 2.95\,x_2 - \int_0^t (x_1(\tau) - r)\,d\tau,$$

with initial condition $\sigma(0) = 0$.